DCIM_Cooling_Prediction Complete Documentation

Folder Structure:

```
dcim_cooling/
│
├── templates/
│   └── index.html
│
├── static/
│   ├── app.js
│   └── style.css
│
├── anomaly.py
├── config.py
├── db.py
├── main.py
├── prediction.py
├── prediction_db.py
└── requirements.txt
```

Path:

10.10.8.22:3020

/opt/nms/dcim_cooling

Execution Flow Inside the Structure:

```
main.py
   ↓
prediction.py
   ↓
db.py (fetch historical data)
   ↓
prediction_db.py (store prediction)
   ↓
anomaly.py (check deviation)
   ↓
index.html (display results)
```

Config.py File:

Config file contain Baseline tolerance in percentage and anomaly threshold. Here the following has been set but can be reconfigured as per user requirement:

ANOMALY_THRESHOLD_KW = 34 # user configurable

TOLERANCE_PERCENT = 0.02

Basic Information About Prediction:

Data taken for training: All data present in database

Features: inlet_temp_c, outlet_temp_c, humidity_pct, it_power_kw, rack_power_kw, cpu_util_pct, fan_speed_pct, outside_temp_c, hour, day_of_week, month.

Model: XGBoost Regression Model (Gradient Boosted Decision Trees)

Trend Followed: The plotted Prediction Data is following its own trend based on the prediction

Baseline: Baseline is the mean of last 24 hours actual value + next 24 hours predicted value

Upper band: Tolerance of 2% from baseline (user config)

threshold: Threshold is set as 34(user config)

Ticketing Logic:

if value>Upper band, show warning in anomaly table

if value>threshold, show warning in anomaly table

**Always set threshold>Upper band

Note: In database we are getting value in 5 mins interval, when we are plotting them, it is 1 hour (mean of last 12 data points) interval to make the graph clean.

Logic behind the predicted data which is Saved in DB (prediction_db.py):

This script runs a background **Cooling prediction service** that executes every 5 minutes or whenever the alert threshold/tolerance changes. It first ensures the prediction_cooling table exists, then checks whether a new prediction cycle is required using a saved state file. If needed, it calls run_prediction(), processes the predicted cooling, baseline, and anomaly flags, and updates the results into the database. Finally, it updates the last run time and configuration state, ensuring controlled and efficient periodic execution.

Service Name:

/etc/systemd/system/cooling_prediction.service

Logs:

/opt/nms/logs/cooling_prediction.log

Report:

Full Anomaly and prediction report is available by clicking **"full report"** button at anomaly table.

Conclusion:

The DCIM Cooling Prediction module:

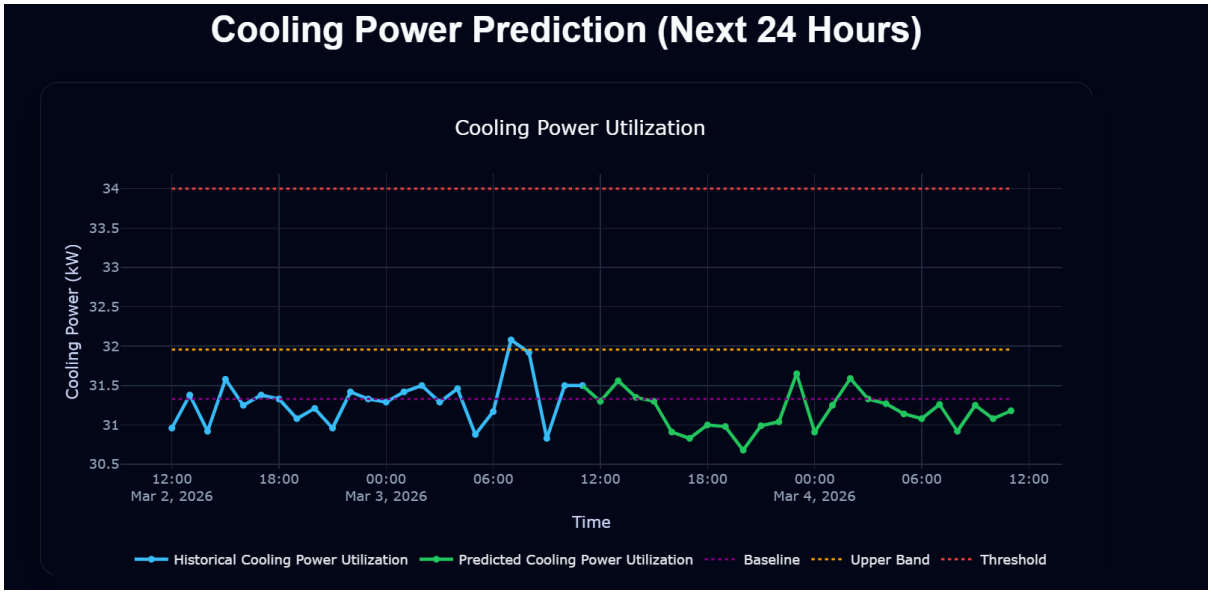
- Predicts future cooling based on historical trends
- Provides anomaly detection
- Supports configurable tolerance & thresholds
- Runs as automated Linux service
- Achieves >95% accuracy

Model Accuracy: Overall, more than 95%

Date	Hour	Cooling Power (kW)	Actual	Efficiency
24-02-2026	17:00	30.73	31.75	96.7874
24-02-2026	18:00	30.79	31.46	97.87031
24-02-2026	19:00	30.66	30.83	99.44859
24-02-2026	20:00	30.72	30.75	99.90244
24-02-2026	21:00	30.23	31.33	96.48899
24-02-2026	22:00	30.32	31.5	96.25397
24-02-2026	23:00	30.75	31.42	97.8676
25-02-2026	00:00	31.13	31	100.4194
25-02-2026	01:00	31.36	30.96	101.292
25-02-2026	02:00	31.44	30.71	102.3771
25-02-2026	03:00	30.9	31	99.67742
25-02-2026	04:00	30.5	31.67	96.30565
25-02-2026	05:00	30.51	31.21	97.75713
25-02-2026	06:00	30.87	31.62	97.62808
25-02-2026	07:00	31.07	31.38	99.01211
25-02-2026	08:00	30.9	30.92	99.93532
25-02-2026	09:00	31.21	31.25	99.872
25-02-2026	10:00	30.47	31.42	96.97645
25-02-2026	11:00	30.46	30.75	99.05691
25-02-2026	12:00	30.11	31.83	94.59629
25-02-2026	13:00	30.31	31.08	97.52252
25-02-2026	14:00	30.18	30.92	97.60673
25-02-2026	15:00	30.43	29.5	103.1525

Screenshots:

Prediction Graph



Prediction Table Download PDF

Date	Hour	Cooling Power (kW)
2026-03-03	12:00	31.30
2026-03-03	13:00	31.56
2026-03-03	14:00	31.35
2026-03-03	15:00	31.30
2026-03-03	16:00	30.91
2026-03-03	17:00	30.83
2026-03-03	18:00	31.00
2026-03-03	19:00	30.98
2026-03-03	20:00	30.68

Anomaly Table Download Report

Date	Hour	Predicted Power (kW)	Expected Baseline (kW)	Upperband	Threshold	Status
No anomalies detected						